NECESSARY AND SUFFICIENT CONDITIONS FOR THE EXISTENCE OF AN LU FACTORIZATION FOR GENERAL RANK DEFICIENT MATRICES*

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Abstract. We establish necessary and sufficient conditions for the existence of an LU factorization $A = LU$ for an arbitrary square matrix A with entries in an arbitrary field F, including singular and rank-deficient cases, without the use of row or column permutations. We prove that such a factorization exists if and only if the nullity of every leading principal submatrix is bounded by the sum of the nullities of the corresponding leading column and row blocks. While building upon the work of Okunev and Johnson, we present simpler, constructive proofs. Furthermore, we extend these results to characterize rank-revealing factorizations, providing explicit sparsity bounds for the factors L and U. Finally, we derive analogous necessary and sufficient conditions for the existence of factorizations constrained to have unit lower or unit upper triangular factors.

Key words. LU factorization, Rank-deficient matrices, Singular matrices, Rank-revealing factorization, Nullity, Sparsity.

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It is well-known that for any square matrix A, there exists a permutation of rows (and potentially columns) such that an LU factorization exists. Specifically, the factorization $PA = LU$, where P is a permutation matrix, L is unit lower triangular, and U is upper triangular, is guaranteed to exist for any matrix A [8, 14].

However, the existence of an LU factorization for a matrix A *without* permutations (that is, $A = LU$) is not guaranteed. A classical result states that for a non-singular matrix, such a factorization exists if and only if all its leading principal minors are non-zero. When the matrix is singular or when leading principal minors vanish, the situation becomes significantly more complex. The standard conditions involving determinants are no longer sufficient to characterize the existence of the factorization.

For these singular and rank-deficient cases, we establish necessary and sufficient conditions for the existence of an LU factorization for an arbitrary matrix A, without imposing restrictions on its rank or invertibility. Throughout, A is a matrix with entries in an arbitrary field F (for example, $\mathbb{R}$, $\mathbb{C}$, the rationals $\mathbb{Q}$). All results hold over any field, since the constructions rely only on rank, nullity, and Gaussian elimination.

The conditions are formulated using the *nullity* of the leading principal submatrices relative to the nullity of the corresponding column and row blocks. Specifically, we prove that an LU factorization exists if and only if for all k (this condition is equivalent to [11]):

$$\mathrm{null}(A[1{:}k, 1{:}k]) \leq \mathrm{null}(A[:, 1{:}k]) + \mathrm{null}(A[1{:}k, :]^T)$$

Note that, in this paper, null is the dimension of the null space of a matrix (see Table 1).

This paper extends the work of Okunev and Johnson [11], who first characterized the existence of an LU factorization by comparing the rank of each leading $k \times k$ principal submatrix with the ranks of its

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first k columns and first k rows. We recast their equivalent condition in nullity form and give a different constructive proof that rests on a restricted form of pivoting: it uses only those internal row or column moves that preserve triangularity once the permutations are undone. The mechanism is an interpretation of a zero pivot through the current Schur complement, where a zero active column corresponds to a globally dependent column, while a zero active row corresponds to a globally dependent row. This yields a direct "restricted pivoting" interpretation of why the existence condition prevents dead ends. The oblique-projection lemmas in the appendix provide an algebraic explanation of why dependent columns or rows become zero in the active Schur complement.

We also develop rank-revealing, sparsity, and unit-triangular consequences from that construction. This manuscript makes the rank profile explicit. It constructs row and column permutations P_r and P_c for a reordered matrix

$$B = P_r^{-1} A P_c^{-1}$$

whose leading $r \times r$ principal blocks are strongly nonsingular, then forms rectangular rank-revealing factors $L_B \in F^{n\times r}$ and $U_B \in F^{r\times n}$. After undoing the internal permutations, the factors $L = P_r L_B$ and $U = U_B P_c$ remain lower and upper triangular. We also state sparsity bounds that connect nonzero structure statements to whether an original row or column is linearly independent of the previous rows or columns.

See also [4] for a discussion on multiple factorizations of singular matrices. Other perspectives on factorizations and their existence in general algebraic structures or specific contexts can be found in [16, 12, 10, 8].

F	An arbitrary field; all matrices have entries in F
$\mathrm{rank}(A)$	Rank of matrix A
$\mathrm{null}(A)$	Dimension of the right null space of matrix A
A^T	Transpose of matrix A
a_{ij}	Entry (i, j) of matrix A
$A[:, i]$	Column i of matrix A
$A[i, :]$	Row i of matrix A
$A[i{:}j, k{:}l]$	Submatrix of A with rows from i to j and columns from k to l
$A[i{:}j, k{:}]$	Submatrix of A with rows from i to j and columns from k to the end
$A[i, k{:}]$	Row i of A, columns from k to the end (and similarly $A[k{:}, j]$ for a column)
e_i	i-th vector of the standard basis
I_n	Identity matrix of size n

Table 1

Notations used in the paper. We use the colon (slicing) notation standard in numerical linear algebra [5]: a colon denotes an index range, and a trailing colon extends the range to the last index, so $A[i, k{:}]$ means columns $k, k+1, \ldots, n$ of row i.

Triangular factorizations and causal systems. Before presenting the technical results, we motivate the study of LU factorizations without permutations by considering their interpretation in causal systems and their computational advantages.

In some applications, the index ordering carries physical significance, such as temporal order in dynamic systems and autoregressive time-series models [7, 6]. A lower triangular matrix L represents a causal operator where the current state i depends only on past states $j \le i$; this is the finite-dimensional matrix analogue of causal discrete-time filtering and of triangular covariance parameterizations for longitudinal data, where coefficients encode regressions on predecessor variables [17, 21]. Similarly, U can be interpreted as capturing reverse-flow dependencies. When $A = LU$, the system respects this intrinsic ordering. However, introducing permutations (as in $PAQ = LU$) destroys this structure: P scrambles the equations (rows) and Q scrambles the variables (columns), breaking the link between the matrix indices and the causal dependencies. Consequently, in contexts where the ordering is immutable (e.g., autoregressive time series, structural equation modeling, causal graphical models, or causal signal processing [1, 19, 20]), standard pivoting strategies are inapplicable. This necessitates a theory for the existence of LU factorizations that strictly preserves the original indexing.

Moreover, the choice of the LU factorization over other decompositions like QR or SVD is often driven by computational efficiency and sparsity. While the QR decomposition always exists for any matrix, the orthogonal factor Q is typically dense. In contrast, the triangular factors L and U often preserve the sparsity of A to a much greater extent, leading to significant savings in storage and computational cost. This cost advantage may also extend to physical limitations, such as data transport or logistical constraints when they are represented by the matrix entries. These reasons lead us to the question of determining exactly when an LU factorization exists without permutations.

We summarize the contributions of this paper, which include:

- A Nullity-based characterization: we prove that the factorization exists if and only if, for every leading principal submatrix, its nullity is at most the sum of the nullities of the corresponding leading column block and leading row block. The proofs are original.
- Rank-revealing structure: we show that we obtain a rank-revealing factorization when the LU factorization exists.
- Sparsity: we provide tight sparsity bounds for the L and U factors.
- Constructive proof: we provide a constructive proof by induction. We include a Python implementation in the appendix to fully characterize the algorithm and allow independent verification of our results.
- Unit triangular factors: we provide a characterization of the conditions for the existence of unit triangular factors.

Organization of paper. The remainder of the paper is organized as follows. Section A reviews classical results regarding factorizations with permutations. Section 1 presents the main theorem and the derivation of the nullity condition. Theorem 1.6 strengthens this result by characterizing rank-revealing factorizations. Finally, Section 2 discusses the specific requirements for unit triangular factors. See Section A for classical results that will serve as reference for the later sections.

1. General necessary and sufficient condition for the existence of an LU factorization.

We address the central question of this paper: determining the necessary and sufficient conditions for the existence of an LU factorization without permutations. We begin with two illustrative examples that motivate the general result.

Consider the matrix:

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

This matrix does not admit an LU factorization. We prove this by contradiction. Suppose $A = LU$. Then the diagonal entry satisfies $a_{11} = l_{11}u_{11} = 0$. This implies that either $l_{11} = 0$ or $u_{11} = 0$.

- If $l_{11} = 0$, then $a_{12} = l_{11}u_{12} = 0$, which contradicts $a_{12} = 1$.
- If $u_{11} = 0$, then $a_{21} = l_{21}u_{11} = 0$, which contradicts $a_{21} = 1$.

In this special case, we have $A^2 = I$ and A is a permutation matrix. So we note that $AA = I \cdot I$ is the factorization of Theorem A.1 with $P = A$, $L = I$, and $U = I$.

However, a zero pivot does not always preclude factorization. Consider the singular matrix:

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Despite the zero principal minors, this matrix *does* have an LU factorization. We can derive it by permuting A into a factorizable form and then reversing the permutations. Apply a cyclic row shift $(1, 2, 3) \to (2, 3, 1)$ via P and swap columns 1 and 3 via Q:

$$PAQ = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

We can recover a factorization for A by writing $A = P^{-1}(PAQ)Q^{-1}$:

$$A = P^{-1} \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \right] Q^{-1} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

This yields a valid, rank-revealing LU factorization with no pivoting. Another valid LU factorization is:

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

We can generalize this result using block matrices. Let 0_n denote the $n \times n$ zero matrix. If B is a factorizable matrix such that $B = LU$, we can construct larger factorizable matrices A as follows:

$$A = \begin{pmatrix} 0_n & 0_n & 0_n \\ 0_n & 0_n & C \\ 0_n & B & D \end{pmatrix} = \begin{pmatrix} 0_n & 0_n \\ C & 0_n \\ D & L \end{pmatrix} \begin{pmatrix} 0_n & 0_n & I_n \\ 0_n & U & 0_n \end{pmatrix} = \begin{pmatrix} 0_n & 0_n \\ CX & 0_n \\ DX & L \end{pmatrix} \begin{pmatrix} 0_n & 0_n & X^{-1} \\ 0_n & U & 0_n \end{pmatrix}$$

for any invertible matrix X.

Alternatively, if $C = LU$:

$$A = \begin{pmatrix} 0_n & 0_n \\ 0_n & L \\ I_n & 0_n \end{pmatrix} \begin{pmatrix} 0_n & B & D \\ 0_n & 0_n & U \end{pmatrix} = \begin{pmatrix} 0_n & 0_n \\ 0_n & L \\ X & 0_n \end{pmatrix} \begin{pmatrix} 0_n & X^{-1}B & X^{-1}D \\ 0_n & 0_n & U \end{pmatrix}$$

These examples demonstrate that the existence of a factorization is linked to the relationship between the ranks (or nullities) of specific sub-blocks. We formalize this observation in Property 1.1. We will show that Property 1.1 below is a necessary and sufficient condition for the existence of an LU factorization.

Property 1.1. *Matrix A is square of size n. For all $k = 1, \dots, n$:*

$$\operatorname{null}(A[1{:}k, 1{:}k]) \leq \operatorname{null}(A[:, 1{:}k]) + \operatorname{null}(A[1{:}k, :]^T) \tag{1.1}$$

We informally describe the main strategy before stating the results formally. The primary obstacle to constructing an LU factorization is the occurrence of a zero pivot. Standard Gaussian elimination resolves this by permuting rows (partial pivoting) or both rows and columns (full pivoting) to bring a non-zero entry to the diagonal.

However, arbitrary permutations destroy the triangular structure required for the specific factorization $A = LU$. If we write the permuted factorization as $PAQ = LU$, recovering A yields $A = (P^{-1}L)(UQ^{-1})$. In general, $P^{-1}L$ is not lower triangular and UQ^{-1} is not upper triangular.

To maintain the triangular factors of A, we must restrict our pivoting strategy. We rely on the observation (detailed in Section D) that a zero column (or row) in the Schur complement corresponds to a column (or row) in the original matrix that is linearly dependent on the preceding columns (or rows).

Our constructive proof effectively "sorts" the matrix indices. Whenever we encounter a zero pivot, the condition in Property 1.1 guarantees that at least one of the following two scenarios must hold:

1. The current column in the Schur complement is zero (the column is linearly dependent). We permute this column rightward.
2. The current row in the Schur complement is zero (the row is linearly dependent). We permute this row downward.

Crucially, because we only permute *zero* columns or rows (relative to the active Schur complement), the triangular structure of the resulting factors is preserved. When we reverse these permutations, the factor $P^{-1}L$ remains lower triangular and UQ^{-1} remains upper triangular.

Condition (1.1) is the precise safeguard that ensures we never reach a "dead end" where the pivot is zero but neither the column nor the row can be safely permuted away (i.e., a case where the rank has not been exhausted, but the leading block is rank-deficient in a way that blocks factorization).

Consequently, this process produces a rank-revealing factorization where the indices are partitioned based on linear independence:

- Linearly Independent Vectors: If a column j_0 of A is linearly independent of the previous columns, it is retained (or moved) into the leading rank-r block ($j \leq r$).
- Linearly Dependent Vectors: If a column j_0 of A is linearly dependent on the previous columns, it is deferred to the trailing block ($j > r$).

A symmetric logic applies to the rows of A.

The main existence results are summarized in Table 2.

Informal description: producer-sensor duality. We can form a mental model of Property 1.1 by considering the following scenario which represents the class of matrices that have an LU factorization. Assume the columns of A correspond to producer points j and the rows of A correspond to sensor points i.

We say the producer j is *observable* by the current sensor network $1{:}i$ if $A[1{:}i, j]$ adds new information (is linearly independent) relative to previous producers $A[1{:}i, 1{:}j-1]$. Conversely, we say the sensor i is

Factorization Type	Equation	Conditions for Existence	Limitations
Non-singular LU	$A = LU$	Strongly non-singular (all leading minors $\neq 0$).	Does not apply to singular matrices.
Partial Pivoting	$PA = LU$	Always exists for any square A.	Permutes rows, altering row-index meaning.
Full Pivoting	$PAQ = LU$	Always exists; rank-revealing.	Permutes rows and columns, destroying all index structure.
Generalized LU	$A = LU$	Subject of this work.	Handles singular/rank-deficient cases without P, Q.

TABLE 2
Summary of the main factorization types and their conditions for existence.

informative regarding the current producers $1{:}j$ if its measurement $A[i, 1{:}j]$ is linearly independent of previous measurements $A[1{:}i-1, 1{:}j]$.

The existence of an LU factorization implies a compatibility between observability and informativeness. A admits a factorization if and only if every "blind spot" (rank deficiency) in a leading block corresponds to a global redundancy. Specifically, if the interaction between the first k sensors and k producers is rank-deficient, this deficiency must be fully accounted for by producers that are globally redundant (dependent on previous columns) or sensors that are globally redundant (dependent on previous rows). The factorization is impossible only in the "crossed" scenario, or "deadlock": where producer k carries a novel signal that current sensors miss (but future sensors see), while sensor k performs a novel measurement that misses current signals (but sees future ones).

We now provide the formal proof of the main results, Theorems 1.5 and 1.6. See Section B for background results on the rank and null space of a matrix and Section C for a proof that the conditions are necessary. These are classical results included for completeness.

1.1. Sufficient condition. The construction below repeatedly removes a zero pivot by interchanging a zero row or column with a later one. We first record a general preservation lemma, then recover the two forms needed below as corollaries.

LEMMA 1.2. *Let A satisfy Property 1.1 with column 1 of A equal to zero, and let $B = A\Pi_j$ interchange columns 1 and j (with $\Pi_j^2 = I$). For $1 \leq k < j$, define the slack*

$$s_k = \operatorname{null}(A[:, 1{:}k]) + \operatorname{null}(A[1{:}k, :]^T) - \operatorname{null}(A[1{:}k, 1{:}k]).$$

Assume that, for every $1 \leq k < j$, if

$$A[:, j] \notin \operatorname{span} A[:, 2{:}k] \quad \text{and} \quad A[1{:}k, j] \in \operatorname{span} A[1{:}k, 2{:}k],$$

then $s_k \geq 1$. Then B satisfies Property 1.1. By transposition, the corresponding statement holds with the roles of rows and columns exchanged.

Proof. For $k \geq j$ the interchange is internal to the first k columns and leaves all three nullities of Property 1.1 unchanged, so the inequality for B reduces to the one for A.

Let $k < j$, and set

$$\epsilon_k = \begin{cases} 1, & A[:,j] \notin \operatorname{span} A[:,2{:}k], \\ 0, & \text{otherwise}, \end{cases} \qquad \delta_k = \begin{cases} 1, & A[1{:}k,j] \notin \operatorname{span} A[1{:}k,2{:}k], \\ 0, & \text{otherwise}. \end{cases}$$

The first k columns of B are obtained from the first k columns of A by replacing the zero column 1 with $A[:,j]$. Hence

$$\operatorname{null}(B[:,1{:}k]) = \operatorname{null}(A[:,1{:}k]) - \epsilon_k.$$

The same argument applied to the leading block gives

$$\operatorname{null}(B[1{:}k,1{:}k]) = \operatorname{null}(A[1{:}k,1{:}k]) - \delta_k.$$

Finally, the interchange only permutes the columns of the row block, so

$$\operatorname{null}(B[1{:}k,:]^T) = \operatorname{null}(A[1{:}k,:]^T).$$

Therefore Property 1.1 for B at index k is equivalent to

$$s_k \geq \epsilon_k - \delta_k.$$

Since A satisfies Property 1.1, we have $s_k \geq 0$. If $\epsilon_k - \delta_k \leq 0$, the inequality follows. If $\epsilon_k - \delta_k = 1$, then $\epsilon_k = 1$ and $\delta_k = 0$, which is exactly the unmatched case covered by the hypothesis. Thus Property 1.1 holds for B for every $k < j$ as well.

The transposed statement follows because Property 1.1 is invariant under transposition: its right-hand side is symmetric in $\operatorname{null}(A[:,1{:}k])$ and $\operatorname{null}(A[1{:}k,:]^T)$, while $\operatorname{null}((A[1{:}k,1{:}k])^T) = \operatorname{null}(A[1{:}k,1{:}k])$. ∎

Corollary 1.3. *Let A satisfy Property 1.1 with column 1 of A equal to zero, and let $B = A\Pi_j$ interchange columns 1 and j (with $\Pi_j^2 = I$). If, for every $k = 1, \dots, j-1$, the truncated column $A[1{:}k,j]$ does not lie in the span of the columns of the leading block $A[1{:}k,1{:}k]$, then B satisfies Property 1.1. By transposition, the corresponding statement holds with the roles of rows and columns exchanged.*

Proof. For each $k < j$, column 1 of A is zero, so the columns of $A[1{:}k,1{:}k]$ span the same space as those of $A[1{:}k,2{:}k]$. Hence the hypothesis says that $A[1{:}k,j] \notin \operatorname{span} A[1{:}k,2{:}k]$ for every $k < j$. The unmatched case in Lemma 1.2 cannot occur, so B satisfies Property 1.1. The row statement follows from the transposed form of Lemma 1.2. ∎

Corollary 1.4. *Let $A \neq 0$ satisfy Property 1.1 with column 1 and row 1 of A equal to zero, and let $B = A\Pi_j$ interchange columns 1 and j (with $\Pi_j^2 = I$). If j is the smallest index whose column is nonzero, then B satisfies Property 1.1. By transposition, the corresponding statement holds with the roles of rows and columns exchanged.*

Proof. For $k < j$, columns $1, \dots, k$ of A all vanish, so

$$\operatorname{null}(A[:,1{:}k]) = \operatorname{null}(A[1{:}k,1{:}k]) = k.$$

Also $\operatorname{null}(A[1{:}k,:]^T) \geq 1$ because row 1 of $A[1{:}k,:]$ is zero. Hence the slack s_k in Lemma 1.2 satisfies $s_k \geq 1$ for every $k < j$, so the hypothesis of that lemma holds and B satisfies Property 1.1. The row statement follows from the transposed form of Lemma 1.2. ∎

We will now prove the main theorem:

THEOREM 1.5. *Matrix A satisfies Property 1.1 if and only if it has an LU factorization.*

Proof. We prove the result by induction on the size of A. The result is true for $n = 1$.

Assume it is true for size $n - 1$.

Assume $a_{11} \neq 0$. Let

$$S = A_{22} - a_{11}^{-1} A_{21} A_{12}$$

be the Schur complement of a_{11} in A. We have the factorization

$$A = LDU$$

where

$$L = \begin{pmatrix} a_{11} & 0 \\ A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} a_{11}^{-1} & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}$$

We show that A satisfies Property 1.1 if and only if D does. Because L is lower triangular and U is upper triangular, for every k the relevant blocks factor as

$$\begin{aligned} A[1{:}k, 1{:}k] &= L[1{:}k, 1{:}k]\, D[1{:}k, 1{:}k]\, U[1{:}k, 1{:}k], \\ A[:, 1{:}k] &= L\, D[:, 1{:}k]\, U[1{:}k, 1{:}k], \\ A[1{:}k, :] &= L[1{:}k, 1{:}k]\, D[1{:}k, :]\, U. \end{aligned}$$

The factors L, U, $L[1{:}k, 1{:}k]$, and $U[1{:}k, 1{:}k]$ are triangular with nonzero diagonal entries (each equal to a_{11} or 1), hence invertible. Since multiplying by an invertible matrix on either side changes neither the rank nor the nullity, we obtain

$$\begin{aligned} \mathrm{null}(A[1{:}k, 1{:}k]) &= \mathrm{null}(D[1{:}k, 1{:}k]), \\ \mathrm{null}(A[:, 1{:}k]) &= \mathrm{null}(D[:, 1{:}k]), \\ \mathrm{null}(A[1{:}k, :]^T) &= \mathrm{null}(D[1{:}k, :]^T) \end{aligned}$$

for every k. Hence the three terms of Property 1.1 coincide for A and for D, which proves the claim. Furthermore, $D = \mathrm{diag}(a_{11}^{-1}, S)$ has a nonsingular leading 1×1 block, so Property 1.1 for D at index k reduces to Property 1.1 for S at index $k-1$; thus D satisfies Property 1.1 if and only if S does. Consequently, S satisfies Property 1.1 and is of size $n - 1$. Moreover S has rank $r - 1$ if $r = \mathrm{rank}(A)$. By the induction hypothesis, S has an LU factorization. The result for A follows.

Assume now $a_{11} = 0$. Evaluating Property 1.1 at $k = 1$ and using $\mathrm{null}(A[1{:}1, 1{:}1]) = 1$,

$$1 \leq \mathrm{null}(A[:, 1]) + \mathrm{null}(A[1{:}1, :]^T), \tag{1.2}$$

so column 1 or row 1 of A is zero. This deadlock-free property is what guarantees the success of our construction below.

If $A = 0$, then $L = U = 0$ is a factorization. Otherwise, let j be the index of the first nonzero column of A and i' the index of the first nonzero row. Transposing A exchanges these two indices and preserves both Property 1.1 and the existence of an LU factorization. We may therefore assume $j \leq i'$.

Then row 1 of A is zero. We prove this by contradiction. Assume it is nonzero, then $i' = 1$, hence $j \leq 1$ and $j = 1$, so column 1 is nonzero. We have a nonzero row and a nonzero column while $a_{11} = 0$. This contradicts (1.2).

Let i now be the smallest index with $A[i,j] \neq 0$. We bring $A[i,j]$ to the $(1,1)$ position by two interchanges, each of a zero line.

- *Column.* If $j > 1$, set $B = A\Pi_j$, interchanging the zero column 1 with column j; by Corollary 1.4, B satisfies Property 1.1. If $j = 1$, set $B = A$. Either way column 1 of B is nonzero and row 1 of B is zero, so $b_{11} = 0$ and $i \geq 2$.
- *Row.* Set $C = \Pi_i B$, interchanging the zero row 1 with row i. Since i is the smallest index with $B[i,1] \neq 0$, column 1 of $B[1{:}k, 1{:}k]$ vanishes for every $k < i$, so the truncated row $B[i, 1{:}k]$ does not lie in the span of the rows of $B[1{:}k, 1{:}k]$; by the row form of Corollary 1.3, C satisfies Property 1.1.

Now $c_{11} = A[i,j] \neq 0$, so by the case already treated C has an LU factorization $C = LU$. The displaced lines of C are zero: row i of C is the old row 1 of A, so row i of L is zero and $\Pi_i L$ is lower triangular; and if $j > 1$, column j of C is the old column 1 of A, so column j of U is zero and $U\Pi_j$ is upper triangular. Since $C = \Pi_i A \Pi_j$,

$$A = \Pi_i \, C \, \Pi_j = (\Pi_i L)(U \Pi_j)$$

is an LU factorization of A.

As in the proof of Theorem A.4, each step reduces the rank of the active matrix by one and the recursion terminates at the zero matrix, so the process performs exactly $r = \operatorname{rank}(A)$ steps. Hence the LU factorization is rank revealing. ∎

We now prove a stronger theorem. Please compare this result with Theorem A.4. The sparsity statements below use the colon (slicing) notation of Table 1 [5]. Recall in particular that a trailing colon denotes a range extending to the last index, so that, for example, $L[i_0, i+1{:}]$ is the part of row i_0 from column $i+1$ onward.

Theorem 1.6. *Let A be a matrix of size n with rank $r \leq n$. If A satisfies Property 1.1, then there exist a row permutation P_r and a column permutation P_c such that the matrix*

$$B = P_r^{-1} A P_c^{-1}$$

has strongly non-singular leading principal submatrices $B[1{:}k, 1{:}k]$ for $1 \leq k \leq r$. Consequently, the leading principal block admits an LU factorization $B[1{:}r, 1{:}r] = L_r U_r$.

Define the rank-revealing factors of B as:

$$L_B = \begin{pmatrix} L_r \\ B[r+1{:}n, 1{:}r] U_r^{-1} \end{pmatrix} \quad \text{and} \quad U_B = \begin{pmatrix} U_r & L_r^{-1} B[1{:}r, r+1{:}n] \end{pmatrix}$$

Then $B = L_B U_B$. Furthermore, A admits a rank-revealing factorization

$$A = LU$$

where $L = P_r L_B$ is lower triangular and $U = U_B P_c$ is upper triangular. We treat L and U as square $n \times n$ factors, obtained by appending $n - r$ zero columns to L_B and $n - r$ zero rows to U_B. Since the factorization is rank revealing, only the first r columns of L and the first r rows of U are populated, so

$$L[:,\, r+1{:}] = 0, \qquad U[r+1{:},\, :] = 0.$$

The remaining entries obey the following sparsity pattern for L. Let i_0 be the row index in A corresponding to the logical row index i in B (i.e., $e_{i_0} = P_r e_i$). Then:

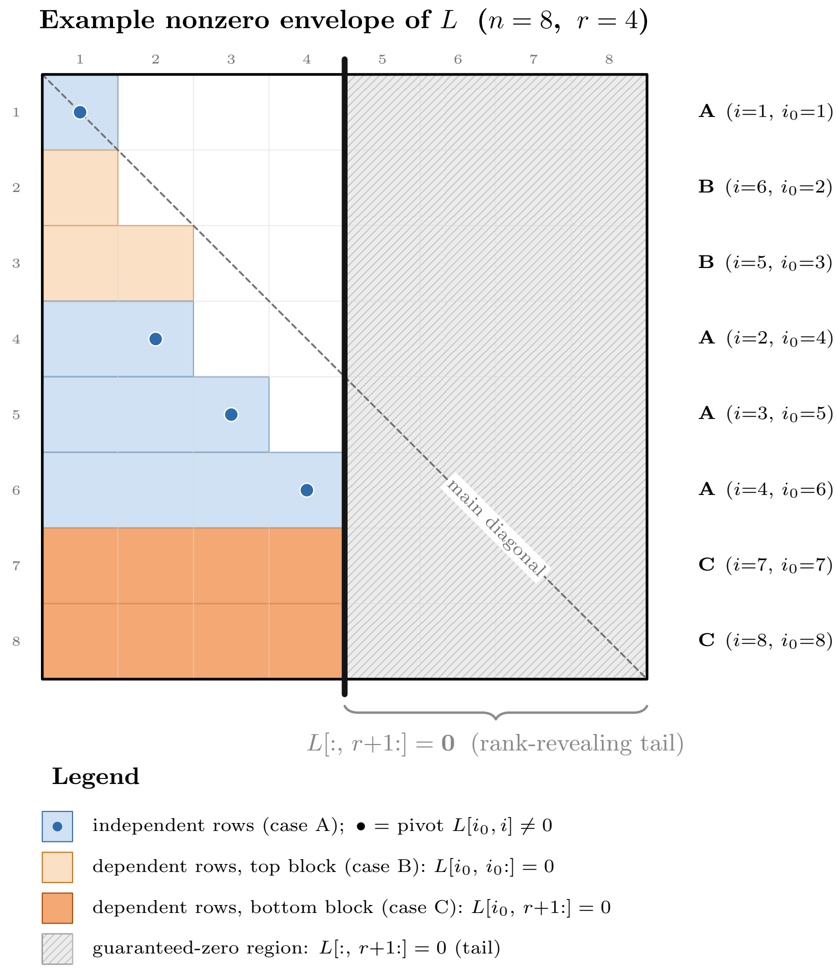

Figure 1. *Figure illustrating the sparsity pattern of the L factor. Case A corresponds to independent rows, $i \leq r$ and $i_0 \geq i$, with $L[i_0, i+1:] = 0$. Case B represents dependent rows, $i > r$ and $i_0 \leq r$. Then $L[i_0, i_0:] = 0$. Case C represents $i > r$ and $i_0 > r$ (which from the proof implies $i_0 = i$) for dependent rows.*

- *Consider $i \leq r$. Then $i_0 \geq i$ and $L[i_0, i+1:] = 0$; in particular, $L[i_0, i_0+1:] = 0$. Row i_0 of A is linearly independent of the previous rows.*
- *Consider $i > r$. If $i_0 > r$, then $i = i_0$. If $i_0 \leq r$, then $L[i_0, i_0:] = 0$. In all cases, row i_0 of A is a linear combination of the previous rows, and $L[i_0, r+1:] = 0$.*

Similarly for U, let j_0 be the column index in A corresponding to the logical column index j in B (i.e., $e_{j_0} = P_c^T e_j$). Then:

- *Consider $j \leq r$. Then $j_0 \geq j$ and $U[j+1:, j_0] = 0$; in particular, $U[j_0+1:, j_0] = 0$. Column j_0 of A is linearly independent of the previous columns.*
- *Consider $j > r$. If $j_0 > r$, then $j = j_0$. If $j_0 \leq r$, then $U[j_0:, j_0] = 0$. In all cases, column j_0 of A is a linear combination of the previous columns, and $U[r+1:, j_0] = 0$.*

Proof. We follow a proof similar to the one for Theorem 1.5.

We do an induction on the size of A. The result is true for $n = 1$.

Assume it is true for size $n-1$. If $a_{11} \neq 0$, let

$$S = A_{22} - A_{21} a_{11}^{-1} A_{12}$$

be the Schur complement of a_{11} in A. We have the factorization

$$A = LDU$$

where

$$L = \begin{pmatrix} a_{11} & 0 \\ A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} a_{11}^{-1} & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}$$

We apply the induction hypothesis to S. We can define P_r and P_c in terms of the permutations for S:

$$P_r = \begin{pmatrix} 1 & 0 \\ 0 & P_{r,S} \end{pmatrix}, \quad P_c = \begin{pmatrix} 1 & 0 \\ 0 & P_{c,S} \end{pmatrix}$$

In this case the first row and first column are the first selected independent row and column. The induction hypothesis gives the stated sparsity pattern for the Schur complement on indices $2, \ldots, n$; adjoining the first pivot only shifts those logical and original indices by one. Since the added first row of the lower factor and first column of the upper factor are triangular, the sparsity claims are preserved.

Assume $a_{11} = 0$. Evaluating Property 1.1 at $k = 1$ gives the deadlock-free alternative used in Theorem 1.5: column 1 or row 1 of A is zero. If $A = 0$, we take $P_r = P_c = I$ and $L = U = 0$, and all sparsity claims are immediate.

Otherwise, let j be the index of the first nonzero column of A and let i' be the index of the first nonzero row of A. Transposing A exchanges the row and column assertions of the theorem, so it is enough to treat the case $j \leq i'$. Then row 1 of A is zero: if it were nonzero, then $i' = 1$, hence $j = 1$, so both row 1 and column 1 would be nonzero, contradicting the deadlock-free alternative at $a_{11} = 0$.

Let i be the smallest index with $A[i,j] \neq 0$. We bring this entry to the $(1,1)$ position exactly as in Theorem 1.5. If $j > 1$, set $B = A\Pi_j$; since column 1 and row 1 of A are zero and j is the first nonzero column, Corollary 1.4 shows that B satisfies Property 1.1. If $j = 1$, set $B = A$. In both cases column 1 of B is nonzero and row 1 of B is zero. Set $C = \Pi_i B$. Since i is the first index with $B[i,1] \neq 0$, the row form of Corollary 1.3 shows that C satisfies Property 1.1, and $c_{11} = A[i,j] \neq 0$.

Applying the nonzero-pivot case to C gives permutations $P_{r,C}$ and $P_{c,C}$ and rank-revealing factors $C = L_C U_C$ with the stated sparsity pattern. Define

$$P_r = \Pi_i P_{r,C}, \qquad P_c = P_{c,C} \Pi_j, \qquad L = \Pi_i L_C, \qquad U = U_C \Pi_j,$$

where $\Pi_j = I$ if $j = 1$. Then

$$B_C = P_{r,C}^{-1} C P_{c,C}^{-1} = P_r^{-1} A P_c^{-1}, \qquad A = LU.$$

The same argument used above for interchanging an identically zero row or column also explains the extra sparsity. Row i of C is the old row 1 of A, hence is zero; because U_C has full row rank on its first r rows, row i of L_C is zero. Thus $\Pi_i L_C$ remains lower triangular, and row 1 of L is zero. If $j > 1$, column j of

C is the old column 1 of A, hence is zero; because L_C has full column rank on its first r columns, column j of U_C is zero. Thus $U_C\Pi_j$ remains upper triangular, and column 1 of U is zero.

Finally, the selected pivot row and column are precisely the next rows and columns that are independent of their predecessors. Indeed, by Lemma D.3, a row or column that is a linear combination of its predecessors becomes zero in the active Schur complement, while an independent one does not. Therefore the first r logical rows and columns are the independent original rows and columns, and the remaining logical rows and columns are dependent. The inequalities $i_0 \geq i$ and $j_0 \geq j$ in the independent cases express that the ith independent row, respectively the jth independent column, cannot occur before original index i, respectively j. In those cases the triangularity of L_C and U_C, transported back by the swaps above, gives $L[i_0, i+1{:}] = 0$ and $U[j+1{:}, j_0] = 0$.

For dependent rows and columns, the zero active row or column is either already beyond the rank-revealing block, in which case it is not moved and $i = i_0$ or $j = j_0$, or it originally occurs at an index at most r and is displaced beyond the first r logical positions. In the latter case its row of L or column of U is zero from the original index onward, giving $L[i_0, i_0{:}] = 0$ or $U[j_0{:}, j_0] = 0$. Since the factorization is rank revealing, the trailing populated parts vanish in all dependent cases:

$$L[i_0, r+1{:}] = 0, \qquad U[r+1{:}, j_0] = 0. \qquad \square$$

1.2. Python implementation. In Subsection E.2, we provide a Python implementation of the algorithm. The algorithm returns L and U such that `A = L @ U`.

Please compare against the algorithm with full pivoting in Subsection E.1, which returns P, Q, L, and U such that `P @ A @ Q = L @ U`. Full pivoting selects the largest pivot for maximum stability. Our algorithm selects the pivot to ensure that the factors L and U remain triangular. We note that magnitude-based pivot selection and the notion of backward stability are specific to ordered or normed fields such as $\mathbb{R}$ and $\mathbb{C}$; over a general field F the existence results still hold, and any nonzero pivot may be selected. The implementations below operate over $\mathbb{R}$ using floating-point arithmetic, but the underlying algorithms only require the field operations of F.

Algorithm 2 is vectorized for performance. The internal permutations are done implicitly. For example, line 34 moves column j to column k. Line 36 zeros out column j of A in exact arithmetic.

This algorithm is not backward stable as we do not control the size of the pivot. Finding a backward stable algorithm will be the topic of future work.

The pseudo-code in [11] (Algorithm 1, p. 12) does not consider the case where the factorization does not exist.

2. Unit triangular factors. We now discuss results where we require one of the factors to be unit triangular. Classical results consider the factorization with partial pivoting $PA = LU$ (Theorem A.1). The row permutation P allows us to obtain a unit lower triangular factor L. When a zero pivot is encountered, we perform a row pivoting to get a non zero pivot. If the entire column is equal to zero, we set the diagonal entry of L to 1, the entries below to 0, and continue the factorization with the next column.

We are interested in conditions for the existence of $A = LU$ where L is unit lower triangular. As before, this requires internal permutations, but this time they can only be applied to the columns. The triangular unit structure of L prevents row permutations. The condition on A is more stringent than in the general LU case and we require

$$\mathrm{null}(A[1{:}k, 1{:}k]) = \mathrm{null}(A[:, 1{:}k])$$

for all k.

We note that, for a general matrix A, column pivoting in general is not sufficient to get a unit lower triangular factor. For example, consider the matrix

$$A = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$$

No column pivoting works. However, row pivoting is sufficient to get a unit lower triangular factor.

Table 3 summarizes the constraints on the factorization, the condition (for all k), and the stringency of the condition.

Constraint	Condition (for all k)	Stringency
General LU	$\mathrm{null}(A_{11}) \le \mathrm{null}(A_{\mathrm{col}}) + \mathrm{null}(A_{\mathrm{row}}^T)$	Least stringent (allows mixed dependencies)
Unit lower L	$\mathrm{null}(A_{11}) = \mathrm{null}(A_{\mathrm{col}})$	Moderate (requires column dependencies)
Unit upper U	$\mathrm{null}(A_{11}) = \mathrm{null}(A_{\mathrm{row}}^T)$	Moderate (requires row dependencies)
Unit L and unit U	A strongly non-singular	Most stringent (impossible for singular A)

Table 3

Constraints on the factorization, the condition (required for all k), and its stringency. Each condition compares the nullities of three submatrices of A: the leading $k \times k$ block $A_{11} = A[1{:}k, 1{:}k]$, the first k columns $A_{\mathrm{col}} = A[:, 1{:}k]$, and the first k rows $A_{\mathrm{row}} = A[1{:}k, :]$. Here $\mathrm{null}(\cdot)$ is the nullity (Table 1), and A_{row}^T is the transpose of the row block, so that $\mathrm{null}(A_{\mathrm{row}}^T) = k - \mathrm{rank}(A_{\mathrm{row}})$.

The theorem below should be compared with Theorem A.1.

Theorem 2.1. *Let A be a matrix of size n and rank $r \le n$. A has an LU factorization $A = LU$ where L is square unit lower triangular if and only if for all $k = 1, \dots, n$:*

$$\mathrm{null}(A[1{:}k, 1{:}k]) = \mathrm{null}(A[:, 1{:}k]) \tag{2.3}$$

Construction: *If A satisfies (2.3), there exists a column permutation P_c such that the matrix $B = AP_c^{-1}$ admits a factorization $B = L_B U_B$ where L_B is unit lower triangular. Consequently,*

$$A = L_B(U_B P_c) = LU$$

is a valid factorization of A with $L = L_B$ unit lower triangular and U upper triangular.

Structure: *For the matrix B constructed above, row j is a linear combination of the previous rows if and only if column j is a linear combination of the previous columns.*

Sparsity: *Let j_0 be the column index in A corresponding to the logical column index j in B (i.e., $e_{j_0} = P_c^T e_j$). The factors exhibit the following sparsity patterns:*

- ***Dependent Case:*** *If row j of B is a linear combination of the previous rows, then $j_0 \le j$. Furthermore:*
$$L[:,j] = e_j, \quad U[j,:] = 0, \quad \text{and} \quad U[j_0:, j_0] = 0$$
- ***Independent Case:*** *If row j of B is linearly independent of the previous rows, then $j_0 \ge j$. Furthermore:*
$$U[j+1:, j_0] = 0$$

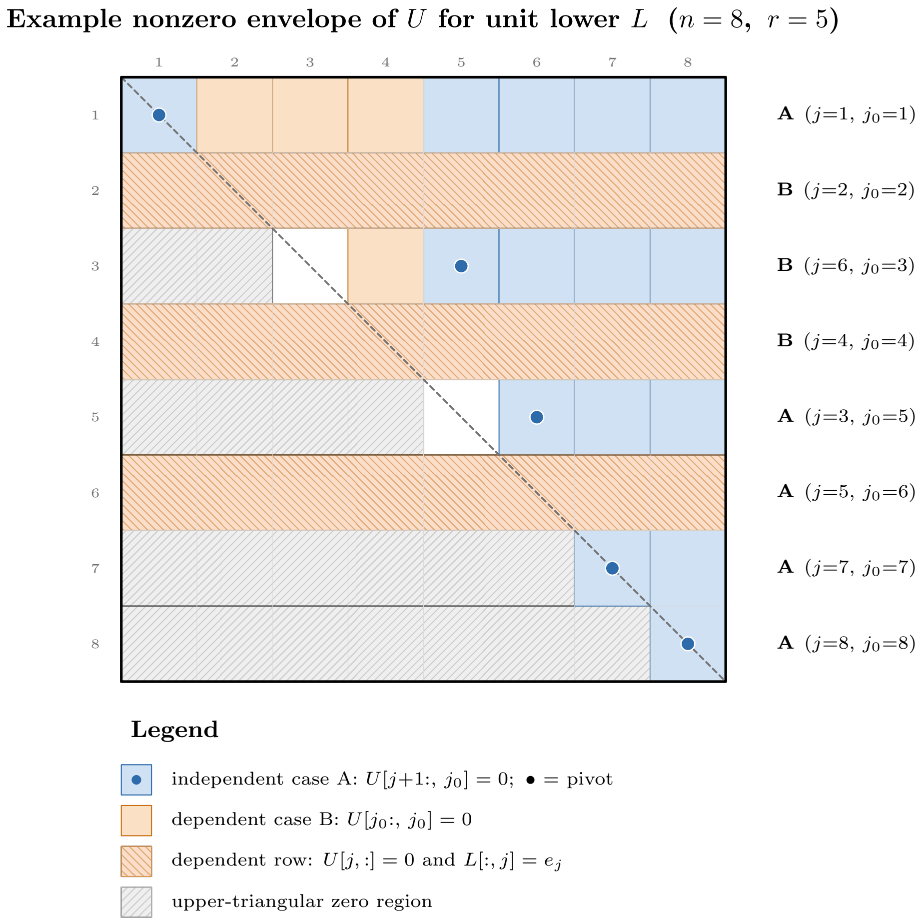

Figure 2. *Figure illustrating the sparsity pattern of the U factor in Theorem 2.1. Case A represents independent rows of B: $j_0 \ge j$ and $U[j+1:, j_0] = 0$. Case B represents dependent rows of B: $j_0 \le j$, $U[j,:] = 0$, and $U[j_0:, j_0] = 0$.*

Proof. **Condition is necessary.** Assume that A admits a factorization $A = LU$, where L is unit lower triangular and U is upper triangular. For any $k \in \{1, \dots, n\}$, since U is upper triangular,

$$A[:, 1{:}k] = L\, U[:, 1{:}k] = L[:, 1{:}k]\, U[1{:}k, 1{:}k],$$
$$A[1{:}k, 1{:}k] = L[1{:}k, 1{:}k]\, U[1{:}k, 1{:}k].$$

Because L is unit lower triangular, $L[1{:}k, 1{:}k]$ is invertible (its determinant is 1), and $L[:, 1{:}k]$ has full column rank (its top block is $L[1{:}k, 1{:}k]$). Multiplying $U[1{:}k, 1{:}k]$ on the left by either matrix therefore leaves its null space unchanged: if $L[:,1{:}k]$ has full column rank, then $L[:,1{:}k]x = 0$ forces $x = 0$, so $A[:,1{:}k]x = 0$ if and only if $U[1{:}k,1{:}k]x = 0$, and likewise for the invertible $L[1{:}k,1{:}k]$. Hence

$$\mathrm{null}(A[1{:}k,1{:}k]) = \mathrm{null}(U[1{:}k,1{:}k]) = \mathrm{null}(A[:,1{:}k]),$$

which is (2.3).

Condition is sufficient. We follow a proof similar to the one for Theorem 1.6. We do an induction on the size of A. The result is true for $n = 1$.

Assume it is true for size $n-1$. If $a_{11} \neq 0$, we verify that the result is true by the induction hypothesis.

Assume $a_{11} = 0$. From (2.3), we have $A[:,1] = 0$.

Case 1: If row 1 of A is non-zero, we can use the same steps as in the proof as Theorem 1.6 with a column permutation to prove the result by induction. In addition, this shows that if $e_{j_0} = P_c^T e_j$ and if row j of B is linearly independent of the previous rows then $j_0 \geq j$ and $U[j+1{:},j_0] = 0$.

Case 2: If row 1 of A is zero, we cannot use the same proof as Theorem 1.6 because we can no longer permute rows. However, since row 1 of A is zero, we can define:

$$A = IDU$$

with

$$D = \begin{pmatrix} 1 & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} 0 & A[1,2{:}n] \\ 0 & I \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & I \end{pmatrix}$$

and $S = A_{22}$. Since row 1 and column 1 of A are zero, for $\ell = 1,\ldots,n-1$,

$$\mathrm{null}(A[1{:}\ell+1,1{:}\ell+1]) = 1 + \mathrm{null}(S[1{:}\ell,1{:}\ell]), \qquad \mathrm{null}(A[:,1{:}\ell+1]) = 1 + \mathrm{null}(S[:,1{:}\ell]).$$

Applying (2.3) to A at $k = \ell + 1$ therefore gives

$$\mathrm{null}(S[1{:}\ell,1{:}\ell]) = \mathrm{null}(S[:,1{:}\ell]), \qquad \ell = 1,\ldots,n-1.$$

Thus S satisfies (2.3). By the induction hypothesis, there is a column permutation $P_{c,S}$ such that

$$B_S = SP_{c,S}^{-1} = L_S U_{B,S},$$

where L_S is unit lower triangular. Set $U_S = U_{B,S}P_{c,S}$, which is upper triangular. Define

$$P_c = \begin{pmatrix} 1 & 0 \\ 0 & P_{c,S} \end{pmatrix}, \quad L_B = \begin{pmatrix} 1 & 0 \\ 0 & L_S \end{pmatrix}, \quad U_B = \begin{pmatrix} 0 & 0 \\ 0 & U_{B,S} \end{pmatrix}.$$

Then $B = AP_c^{-1} = L_B U_B$, and

$$A = L_B(U_B P_c), \qquad U_B P_c = \begin{pmatrix} 0 & 0 \\ 0 & U_S \end{pmatrix}$$

is upper triangular. We set $L = L_B$ and $U = U_B P_c$.

The structure and sparsity statements in this second case are the induction statements for S, shifted by one index, together with the new zero first row and column. Row 1 and column 1 of B are both zero, so they are both dependent, $j_0 = 1$, and $L[:,1] = e_1$, $U[1,:] = 0$, $U[1:,1] = 0$. For $j > 1$, write $\ell = j - 1$ and $\ell_0 = j_0 - 1$. Dependence of row j of B on the previous rows is equivalent to dependence of row ℓ of B_S on the previous rows. The same statement holds for columns. Hence the induction hypothesis for S gives the equivalence between dependent rows and dependent columns for B. It also gives, in the dependent case, $\ell_0 \leq \ell$, $L_S[:,\ell] = e_\ell$, $U_S[\ell,:] = 0$, and $U_S[\ell_0:,\ell_0] = 0$. Shifting the indices yields $j_0 \leq j$, $L[:,j] = e_j$, $U[j,:] = 0$, and $U[j_0:,j_0] = 0$. In the independent case, the induction hypothesis gives $\ell_0 \geq \ell$ and $U_S[\ell+1:,\ell_0] = 0$, which shifts to $j_0 \geq j$ and $U[j+1:,j_0] = 0$.

Note that the step above with D is not a valid step in Theorem 1.6. Indeed:

$$\begin{aligned}
\mathrm{null}(D[1{:}k, 1{:}k]) &= \mathrm{null}(A[1{:}k, 1{:}k]) - 1 \\
\mathrm{null}(D[:, 1{:}k]) &= \mathrm{null}(A[:, 1{:}k]) - 1 \\
\mathrm{null}(D[1{:}k, :]^T) &= \mathrm{null}(A[1{:}k, :]^T) - 1
\end{aligned}$$

∎

We now state the corresponding result where U is unit upper triangular. Please compare the result below with Corollary A.2.

COROLLARY 2.2. *Let A be a matrix of size n and rank $r \leq n$. A has an LU factorization $A = LU$ where U is unit upper triangular if and only if for all $k = 1, \ldots, n$:*

$$\mathrm{null}(A[1{:}k, 1{:}k]) = \mathrm{null}(A[1{:}k, :]^T) \tag{2.4}$$

Construction: *If A satisfies (2.4), there exists a row permutation P_r such that the matrix $B = P_r^{-1}A$ admits a factorization $B = L_B U_B$ where U_B is unit upper triangular. Consequently,*

$$A = (P_r L_B)U_B = LU$$

is a valid factorization of A with $U = U_B$ unit upper triangular and L lower triangular.

Structure: *For the matrix B constructed above, column i is a linear combination of the previous columns if and only if row i is a linear combination of the previous rows.*

Sparsity: *Let i_0 be the row index in A corresponding to the logical row index i in B (i.e., $e_{i_0} = P_r e_i$). The factors exhibit the following sparsity patterns:*

- ***Dependent Case:*** *If column i of B is a linear combination of the previous columns, then $i_0 \leq i$. Furthermore:*

$$U[i,:] = e_i, \quad L[:,i] = 0, \quad \textit{and} \quad L[i_0, i_0{:}] = 0$$

- ***Independent Case:*** *If column i of B is linearly independent of the previous columns, then $i_0 \geq i$. Furthermore:*

$$L[i_0, i+1{:}] = 0$$

Proof. Consider $B = A^T$. Apply Theorem 2.1. ∎

2.1. Python implementation. In Subsection E.3, we provide a Python implementation of the algorithm. This code is significantly simpler since we only do internal permutations on the columns of A. As previously, the internal permutations are done implicitly, and this algorithm is not backward stable.

3. Conclusion. In this paper, we have provided an analysis of the necessary and sufficient conditions for the existence of an LU factorization for arbitrary square matrices, without the use of pivoting. We established that the existence of such a factorization is determined by a specific inequality relating the nullity of the leading principal submatrices to the nullity of the corresponding leading column and row blocks. This condition generalizes the classical requirement for non-singular matrices (non-vanishing leading principal minors) to the general rank-deficient case.

Our approach utilizes a constructive proof based on induction. This method not only simplifies the theoretical derivation but also provides explicit insight into how singular blocks can be handled through internal permutations that preserve the overall triangular structure of the factors.

Furthermore, we defined and characterized rank-revealing LU factorizations without global pivoting. We derived tight sparsity bounds for the resulting factors L and U, linking the physical indices of the non-zero entries to the logical rank profile of the matrix. Finally, we examined the constrained cases where one factor is required to be unit triangular, identifying the specific rank conditions required and their duality with respect to row and column permutations. These results offer a unified framework for understanding matrix factorizations in the presence of singularity and rank deficiency.

Appendix A. Classical results on the existence of LU factorizations.

We recall standard existence results involving permutations. These theorems correspond to Gaussian elimination with partial and full pivoting.

We work over the arbitrary field F. Since F need not carry an inner product, we use Y^T purely as the transpose, and for a matrix $Y \in F^{n\times k}$ we write

$$Y^\perp = \ker(Y^T) = \{\, v \in F^n : Y^T v = 0 \,\}.$$

This is a purely algebraic construction (the annihilator of the row space under the standard bilinear pairing $x^T y = \sum_i x_i y_i$) and is defined over any field; over $\mathbb{R}$ it coincides with the usual orthogonal complement.

Theorem A.1 (Partial Pivoting). *For any square matrix A of size n, there exists a permutation matrix P such that $PA = LU$, where L is unit lower triangular and U is upper triangular.*

Proof. This is a classical result. We prove it by induction on n. The result is true for $n = 1$. Assume it is true for matrices of size $n-1$, and write A in block form with leading 1×1 block a_{11}:

$$A = \begin{pmatrix} a_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}.$$

Case 1: $a_{11} \neq 0$. Define the Schur complement of a_{11} in A,

$$S = A_{22} - A_{21} a_{11}^{-1} A_{12},$$

which is a matrix of size $n-1$. A direct computation gives the block factorization $A = LDU$ with

$$L = \begin{pmatrix} 1 & 0 \\ a_{11}^{-1} A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}.$$

By the induction hypothesis applied to S, there exists a permutation matrix P_s such that $P_s S = L_s U_s$, with L_s unit lower triangular and U_s upper triangular. This row permutation of S extends to a row permutation of A that fixes the first row, namely

$$\widehat{P}_s = \begin{pmatrix} 1 & 0 \\ 0 & P_s \end{pmatrix},$$

which is again a permutation matrix. Using $A = LDU$ and then $P_s S = L_s U_s$, we obtain

$$\widehat{P}_s A = \begin{pmatrix} 1 & 0 \\ a_{11}^{-1} P_s A_{21} & P_s \end{pmatrix} \begin{pmatrix} a_{11} & A_{12} \\ 0 & S \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ a_{11}^{-1} P_s A_{21} & L_s \end{pmatrix} \begin{pmatrix} a_{11} & A_{12} \\ 0 & U_s \end{pmatrix}.$$

The last product is an LU factorization: the left factor is unit lower triangular and the right factor is upper triangular. Hence $P = \widehat{P}_s$ proves the result for A.

Case 2: $a_{11} = 0$ and column 1 of A is non-zero. Let i be the smallest index such that $A[i, 1] \neq 0$, and let Π_i be the permutation that swaps rows 1 and i (so $\Pi_i^2 = I$). The matrix $B = \Pi_i A$ has $b_{11} = A[i, 1] \neq 0$, so by Case 1 there exists a permutation matrix $\widehat{P}$ with $\widehat{P} B = LU$. Then $(\widehat{P}\Pi_i)A = LU$, and $P = \widehat{P}\Pi_i$ proves the result for A.

Case 3: $a_{11} = 0$ and column 1 of A is zero. Then $A_{21} = 0$, and setting $S = A_{22}$ gives the factorization $A = IDU$ with

$$D = \begin{pmatrix} 1 & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} 0 & A_{12} \\ 0 & I \end{pmatrix}.$$

By the induction hypothesis applied to S, there exists a permutation matrix P_s such that $P_s S = L_s U_s$. With $\widehat{P}_s = \operatorname{diag}(1, P_s)$ as in Case 1,

$$\widehat{P}_s A = \begin{pmatrix} 0 & A_{12} \\ 0 & P_s S \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & L_s \end{pmatrix} \begin{pmatrix} 0 & A_{12} \\ 0 & U_s \end{pmatrix},$$

which is an LU factorization. Hence $P = \widehat{P}_s$ proves the result for A. ∎

Corollary A.2. *For any square matrix A of size n, there exists a permutation matrix Q such that $AQ = LU$, where L is lower triangular and U is unit upper triangular.*

Proof. Apply Theorem A.1 to A^T. ∎

Definition A.3. *A matrix A of size $m \times n$ is said to have a* rank revealing LU factorization *if it can be written as $A = LU$, where L is a lower triangular matrix of size $m \times r$, U is an upper triangular matrix of size $r \times n$, and $r = \operatorname{rank}(A)$.*

If we allow both row and column permutations, we can always obtain such a factorization.

Theorem A.4 (Full Pivoting). *For any matrix A of size $m \times n$, there exist a row permutation matrix P and a column permutation matrix Q such that $PAQ = LU$ is a rank revealing LU factorization.*

Proof. We use induction on n. The result is true for $n = 1$. Assume it holds for matrices with $n - 1$ columns, and write A in block form with leading 1×1 block a_{11}:

$$A = \begin{pmatrix} a_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}.$$

Case 1: $a_{11} \neq 0$. Define the Schur complement of a_{11} in A,

$$S = A_{22} - A_{21} a_{11}^{-1} A_{12},$$

a matrix of size $(m-1)\times(n-1)$. A direct computation gives the block factorization $A = LDU$ with

$$L = \begin{pmatrix} a_{11} & 0 \\ A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} a_{11}^{-1} & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}.$$

Since L and U are square and invertible, $\mathrm{rank}(A) = 1 + \mathrm{rank}(S)$; hence S has rank $r-1$, where $r = \mathrm{rank}(A)$. By the induction hypothesis applied to S, there exist a row permutation P_s and a column permutation Q_s such that $P_s S Q_s = L_s U_s$ is rank revealing, with L_s of size $(m-1)\times(r-1)$ lower triangular and U_s of size $(r-1)\times(n-1)$ upper triangular. These permutations of S extend to permutations of A that fix the first row and the first column,

$$\widehat{P}_s = \begin{pmatrix} 1 & 0 \\ 0 & P_s \end{pmatrix}, \quad \widehat{Q}_s = \begin{pmatrix} 1 & 0 \\ 0 & Q_s \end{pmatrix},$$

which are again permutation matrices. Using $A = LDU$ and then $P_s S Q_s = L_s U_s$, we obtain

$$\widehat{P}_s A \widehat{Q}_s = \begin{pmatrix} a_{11} & 0 \\ P_s A_{21} & I \end{pmatrix} \begin{pmatrix} 1 & a_{11}^{-1} A_{12} Q_s \\ 0 & P_s S Q_s \end{pmatrix} = \begin{pmatrix} a_{11} & 0 \\ P_s A_{21} & L_s \end{pmatrix} \begin{pmatrix} 1 & a_{11}^{-1} A_{12} Q_s \\ 0 & U_s \end{pmatrix}.$$

The left factor is of size $m \times r$ and lower triangular, the right factor is of size $r \times n$ and upper triangular, so this is a rank revealing LU factorization. Hence $P = \widehat{P}_s$ and $Q = \widehat{Q}_s$ prove the result for A.

Case 2: $a_{11} = 0$. If $A = 0$, then $r = 0$ and the empty factorization $L = U = 0$ works. Otherwise, choose i and j with $a_{ij} \neq 0$, and let Π_i and Π_j be the permutations swapping rows $1, i$ and columns $1, j$, respectively. The matrix $B = \Pi_i A \Pi_j$ has $b_{11} = a_{ij} \neq 0$, so by Case 1 there exist permutations $\widehat{P}$ and $\widehat{Q}$ such that $\widehat{P} B \widehat{Q} = LU$ is rank revealing. Then $(\widehat{P}\Pi_i) A (\Pi_j \widehat{Q}) = LU$, and $P = \widehat{P}\Pi_i$, $Q = \Pi_j \widehat{Q}$ prove the result for A.

This recursion terminates after exactly $r = \mathrm{rank}(A)$ steps. Indeed, each step in Case 1 passes from A to its Schur complement S, which has rank $r-1$, so the rank of the active matrix drops by exactly one per pivot; and the recursion bottoms out at the $A = 0$ subcase of Case 2, which is reached precisely when the rank reaches 0. Starting from rank r, the process therefore performs r pivot steps, and the resulting factorization has inner dimension r; that is, it is rank revealing. ∎

We now consider LU factorizations without permutations. The fundamental result states that factorization is possible if and only if the matrix is *strongly non-singular* (i.e., all leading principal minors are non-zero).

PROPOSITION A.5. *Let A be a non-singular matrix of size n. A has an LU factorization if and only if all its leading principal submatrices are non-singular:*

$$\mathrm{rank}(A[1{:}k, 1{:}k]) = k, \quad k = 1, \ldots, n$$

Proof. We prove that the condition is necessary. Assume $A = LU$:

$$\det(A) = \prod_{i=1}^{n} l_{ii} u_{ii}$$

and $\det(A) \neq 0$. So, $l_{ii} \neq 0$ and $u_{ii} \neq 0$ for all i. Therefore, $\mathrm{rank}(A[1{:}k, 1{:}k]) = k$, $k = 1, \ldots, n$.

We prove that the condition is sufficient. We prove the result by induction on n. The result is true for $n = 1$.

Assume it is true for $n-1$. By assumption, $a_{11} \neq 0$. We then have the factorization:

$$A = LDU$$

where

$$L = \begin{pmatrix} a_{11} & 0 \\ A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} a_{11}^{-1} & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}$$

and $S = A_{22} - A_{21}a_{11}^{-1}A_{12}$ is the Schur complement. L and U are full rank. So D satisfies the condition of the theorem. Consequently, S also satisfies the condition of the theorem and is of size $n-1$. By the induction hypothesis, S has an LU factorization. Therefore A also has an LU factorization. ∎

Appendix B. Rank theorems.

We recall rank theorems that will be useful in the proof of Theorem 1.5.

THEOREM B.1. *The rank-nullity theorem states that for any matrix A of size $m \times n$:*

$$\operatorname{rank}(A) + \operatorname{null}(A) = n$$

We now state Sylvester's inequality for the rank of a product of matrices.

LEMMA B.2. *Consider $A = BC$. Then:*

$$\operatorname{rank}(A) \leq \min(\operatorname{rank}(B), \operatorname{rank}(C)), \quad \operatorname{rank}(B) + \operatorname{rank}(C) - k \leq \operatorname{rank}(A)$$

where k is the number of columns of B and rows of C.

COROLLARY B.3. *If A, B, and C are square matrices then:*

$$\operatorname{null}(A) \leq \operatorname{null}(B) + \operatorname{null}(C)$$

Proof. Assume $A = BC$ where A, B, C are $n \times n$ matrices. From Lemma B.2:

$$\operatorname{rank}(A) \geq \operatorname{rank}(B) + \operatorname{rank}(C) - n$$

Using Theorem B.1, $\operatorname{rank}(M) = n - \operatorname{null}(M)$:

$$n - \operatorname{null}(A) \geq (n - \operatorname{null}(B)) + (n - \operatorname{null}(C)) - n$$

Simplifying: $\operatorname{null}(A) \leq \operatorname{null}(B) + \operatorname{null}(C)$. ∎

Appendix C. Necessary condition.

We show that Property 1.1 is a necessary condition for the existence of an LU factorization.

Assume that A, a square matrix of size n, has an LU factorization:

$$A = LU$$

Then:

$$\begin{aligned} A[1{:}k, :] &= L[1{:}k, 1{:}k]U[1{:}k, :], \\ A[:, 1{:}k] &= L[:, 1{:}k]U[1{:}k, 1{:}k], \\ A[1{:}k, 1{:}k] &= L[1{:}k, 1{:}k]U[1{:}k, 1{:}k] \end{aligned}$$

Using Lemma B.2 and Corollary B.3, we have:

$$\begin{aligned}\operatorname{rank}(A[1{:}k,:]) &\le \operatorname{rank}(L[1{:}k,1{:}k]),\\ \operatorname{rank}(A[:,1{:}k]) &\le \operatorname{rank}(U[1{:}k,1{:}k]),\\ \operatorname{null}(A[1{:}k,1{:}k]) &\le \operatorname{null}(L[1{:}k,1{:}k]) + \operatorname{null}(U[1{:}k,1{:}k])\end{aligned}$$

Using Theorem B.1, we get:

$$\operatorname{null}(A[1{:}k,1{:}k]) \le \operatorname{null}(A[:,1{:}k]) + \operatorname{null}(A[1{:}k,:]^T)$$

So Property 1.1 is a necessary condition for the existence of an LU factorization.

Appendix D. Oblique projections.

We recall a few standard results, included to keep the paper self-contained, that are used in the main proofs. This material is classical: the oblique projectors below are treated in [22, 13], and their connection to the LU factorization and Schur complements in [3, 18, 23, 9].

Lemma D.1. *Denote by X and Y two matrices of size $n \times k$. Assume that X is full rank and that $Y^T X$ is non-singular. Define:*

$$P = X(Y^T X)^{-1} Y^T$$

Then P is the oblique projection onto the range of X along $Y^\perp = \ker(Y^T)$. In particular, $\operatorname{range}(X) \oplus \ker(Y^T) = F^n$.

Proof. We have:

$$PX = X(Y^T X)^{-1} Y^T X = X$$

So the range of P contains the range of X. Since $\operatorname{rank}(P) \le k$ and $\operatorname{rank}(X) = k$, the range of P is exactly the range of X.

Consider v in $Y^\perp = \ker(Y^T)$ (i.e., $Y^T v = 0$). We have:

$$Pv = X(Y^T X)^{-1} Y^T v = 0$$

Thus, the null space of P contains $\ker(Y^T)$. By the rank-nullity theorem, the dimensions match, so the null space of P is exactly $\ker(Y^T)$. ∎

Lemma D.2. *Denote by $X = [x_1, \ldots, x_k]$ a rank k matrix. Consider the oblique projection $P_o(X)$ onto*

$$S = \operatorname{span}\{x_1, \ldots, x_k\}$$

along the subspace of vectors with the first k entries equal to zero. Assume that $X[1{:}k,:]$ is non-singular. Then:

$$P_o(X) = X(X[1{:}k,:])^{-1} Y^T$$

where

$$Y = \begin{pmatrix} I_k \\ 0 \end{pmatrix}$$

Proof. In Lemma D.1, set Y as

$$Y = \begin{pmatrix} I_k \\ 0 \end{pmatrix}$$

The range of Y is the span of the first k standard basis vectors. The subspace $\ker(Y^T)$ is exactly the set of vectors with the first k entries equal to zero. ∎

Denote by $Q_o(X) = I - P_o(X)$ the oblique projection onto the subspace of vectors with the first k entries equal to zero along $S = \mathrm{span}(X)$.

Consider the state of the matrix after k steps of the LU factorization algorithm (the first k columns have been eliminated). The Schur complement can be interpreted in terms of the oblique projection Q_o. Specifically, from Lemma D.2 applied to the first k columns, we have:

$$S_k^o = A[:, k+1:] - A[:, 1:k](A[1:k, 1:k])^{-1} A[1:k, k+1:] = Q_o(A[:, 1:k])\ A[:, k+1:]$$

The standard Schur complement S_k, a matrix of size $n-k$, corresponds to the bottom block of this projected matrix:

$$S_k = S_k^o[k+1:, :].$$

Applying $Q_o(A[:, 1:k]) = I - P_o(A[:, 1:k])$ to all of A, rather than only to its trailing columns, gives

$$(I - P_o(A[:, 1:k]))\, A = \begin{pmatrix} 0_k & 0 \\ 0 & S_k \end{pmatrix},$$

where 0_k is the $k \times k$ zero block.

This identity makes the link between the projection and Gaussian elimination explicit: the single factor $Q_o(A[:, 1:k])$ realizes the first k elimination steps. Its range is the subspace of vectors whose first k entries are zero, so every column of $Q_o(A[:, 1:k])\, A$ has zero first k entries; moreover, the first k columns of A are annihilated because they span the range of $P_o(A[:, 1:k])$. The first k rows and columns therefore vanish, leaving the $(n-k) \times (n-k)$ Schur complement S_k on which the elimination continues.

Lemma D.3. *If column j of A (where $j > k$) is linearly dependent on columns 1 to k of A, then column $j - k$ of S_k^o is zero.*

Proof. Since column j of A is linearly dependent on columns 1 to k, we have:

$$A[:, j] \in \mathrm{span}\{A[:, 1], \ldots, A[:, k]\}$$

Let $X = A[:, 1:k]$. Then $A[:, j] \in \mathrm{span}(X)$. By definition of the projection $P_o(X)$:

$$P_o(X)A[:, j] = A[:, j]$$

Therefore:

$$Q_o(X)A[:, j] = (I - P_o(X))A[:, j] = 0$$

Since S_k^o is formed by applying $Q_o(X)$ to the columns $A[:, k+1:]$, the column corresponding to $A[:, j]$ is zero. ∎

The vanishing of dependent columns in the Schur complement is a classical property. When $A[1:k, 1:k]$ is non-singular, it underlies the rank formula $\mathrm{rank}(A) = \mathrm{rank}(A[1:k, 1:k]) + \mathrm{rank}(S_k)$; see [15, 2, 3].

Appendix E. Reference Python code.

E.1. Classical LU with full pivoting.

See Algorithm 1.

```python
def lu_pivoting(A_in):
    A = A_in.astype(np.float64).copy()
    n = A.shape[0]
    L, U = np.zeros_like(A), np.zeros_like(A)
    p, q = np.arange(n), np.arange(n)

    def swap_rows(k, i):
        A[[k, i], k:] = A[[i, k], k:]
        p[[k, i]] = p[[i, k]]
        L[[k, i], :k] = L[[i, k], :k]

    def swap_cols(k, j):
        A[k:, [k, j]] = A[k:, [j, k]]
        q[[k, j]] = q[[j, k]]
        U[:k, [k, j]] = U[:k, [j, k]]

    for k in range(n):
        idx_max = np.argmax(np.abs(A[k:, k:]))
        i_local, j_local = np.unravel_index(idx_max, A[k:, k:].shape)
        i_max, j_max = k + i_local, k + j_local
        swap_rows(k, i_max)
        swap_cols(k, j_max)
        if np.isclose(A[k, k], 0.0): break

        L[k:, k] = A[k:, k] / A[k, k]
        U[k, k:] = A[k, k:]
        A[k+1:, k+1:] -= np.outer(L[k+1:, k], U[k, k+1:])

    P, Q = np.eye(n)[p], np.eye(n)[:, q]
    return P, Q, L, U
```

Algorithm 1

Classical LU factorization with full pivoting. Returns a factorization of the form $PAQ = LU$.

E.2. LU factorization. See Algorithm 2.

```python
def lu_factorization(A_in):
    A = A_in.astype(np.float64).copy()
    n = A.shape[0]
    L, U = np.zeros_like(A), np.zeros_like(A)
    for k in range(n):
        i, j = k, k
        if np.isclose(A[i, j], 0.0):
            # Search for the first shell ‘i‘ (relative to k) that contains a non-zero element in either
                its row or column slice.
            is_nz = ~np.isclose(A[k:, k:], 0.0)

            # valid_rows[x] is True if Row x of A[k:, k:] (upper part, j>=x) has non-zero
            valid_rows = np.any(np.triu(is_nz), axis=1)
            # valid_cols[x] is True if Col x of A[k:, k:] (lower part, i>=x) has non-zero
            valid_cols = np.any(np.tril(is_nz), axis=0)
            valid_shells = valid_rows | valid_cols

            if np.any(valid_shells):
                l_rel = np.argmax(valid_shells) # First index where valid
                l = k + l_rel
                # Find k_row and k_col for this specific shell l
                k_row = l + np.argmax(~np.isclose(A[l, l:], 0.0)) if valid_rows[l_rel] else n + 1
                k_col = l + np.argmax(~np.isclose(A[l:, l], 0.0)) if valid_cols[l_rel] else n + 1
                use_row = k_row <= k_col
                if l == k and np.any(~np.isclose(A[k:, k] if use_row else A[k, k:], 0.0)):
                    # Factorization does not exist
                    return None
                i, j = (l, k_row) if use_row else (k_col, l)
            else:
                break

```

```
        L[k:, k] = A[k:, j] / A[i, j] #
        U[k, k:] = A[i, k:]
        A[k+1:, k+1:] -= np.outer(L[k+1:, k], U[k, k+1:]) #

    return L, U
```

Algorithm 2

Algorithm for LU factorization. Returns the LU factorization of a matrix $A = LU$ or None if the factorization does not exist. The algorithm is guaranteed to return None only when the factorization does not exist.

E.3. Unit lower triangular LU factorization. See Algorithm 3.

```
def unit_lower_lu_factorization(A_in):
    A = A_in.astype(np.float64).copy()
    n = A.shape[0]
    L, U = np.eye(n), np.zeros_like(A)

    for k in range(n):
        j = k + np.argmax(~np.isclose(A[k, k:], 0.0))
        if np.isclose(A[k, k], 0.0):
            if not np.allclose(A[k+1:, k], 0.0):
                # Factorization does not exist
                return None
            if np.isclose(A[k, j], 0.0): continue

        L[k+1:, k] = A[k+1:, j] / A[k, j]
        U[k, k:] = A[k, k:]
        A[k+1:, k+1:] -= np.outer(L[k+1:, k], U[k, k+1:])

    return L, U
```

Algorithm 3

Unit lower triangular LU factorization. Returns a factorization $A = LU$ where L is unit lower triangular or None if the factorization does not exist.

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